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Deep learning-based seed germination prediction using morphological traits and RGB images

Olçay Tosun^{1,2*} and Recep Eryiğit^{1*}

Abstract

Seed selection constitutes the initial and one of the most critical steps in agricultural productivity. The identification of high-quality seeds is a labor-intensive and costly process that requires considerable expertise. Within the scope of smart farming applications, this study proposes a deep learning-based model designed to automate the seed selection process by accurately predicting seed germination capacity from seed images. The proposed model determines whether a seed will germinate using RGB images and morphological traits automatically extracted from these images. The dataset used in this study comprises a total of 3,645 images belonging to three different seed types. For each seed type (okra, eggplant, and tomato), 405 seed images were acquired from three distinct imaging sources (digital microscope, camera, and scanner), labeled, and subsequently sown in seed trays. The germination status of each sown seed was systematically monitored and matched with its corresponding image data. The dataset was partitioned into 80% training and 20% testing subsets. Following 5-fold stratified cross-validation, the proposed model achieved an average weighted F1-score of 0.95 on the training set and 0.93 on the testing set for germination capacity prediction. The performance of the proposed model was further compared with widely used deep convolutional neural network architectures, including VGG19, ResNet50, and EfficientNetB5. Comparative results demonstrate that the proposed model provides competitive and robust performance for seed germination prediction. Overall, the findings indicate that the proposed approach can effectively be utilized for automated seed germination prediction. Future research should evaluate the generalizability of the model by conducting performance assessments on additional seed types.

Keywords seed germination, deep learning, CNN, prediction, smart agriculture

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Introduction

Sustainable development, as defined in the literature, aims to meet present needs without compromising the ability of future generations to meet their own needs. In line with this objective, international organizations and national governments have developed and implemented policies to promote long-term environmental integrity, social equity, and economic stability. These three inter-related pillars (environmental, social, and economic dimensions) constitute the core framework of sustainable development. Within this framework, sustainable agriculture plays a critical role in maintaining balance among these dimensions. It aims not only to meet current food, and textile demands but also to safeguard the productive capacity of agricultural systems for future generations. Beyond its contribution to food security, agriculture represents a significant component of national and global economies. According to World Trade Organization (WTO) data, the global trade value of agricultural products increased substantially between 2000 and 2023, with imports rising from USD 325 billion to USD 1,521 billion and exports from USD 300 billion to USD 1,465 billion. During this period, exports of agricultural commodities demonstrated remarkable growth, including increases of 557% in oilseeds and oils and 370% in grains [1].

The primary objective in crop production is to achieve higher yields with fewer inputs, particularly given that plant-based production accounts for more than 80% of global food consumption. Achieving this objective requires careful optimization of key agricultural inputs, among which seed selection represents the most fundamental and decisive factor. As the initial biological input in crop production systems, seed quality directly influences productivity, resource efficiency, and overall sustainability. In recent years, artificial intelligence (AI) applications have been increasingly developed and integrated into various stages of agricultural production, ranging from seed selection to post-harvest processes and distribution. In particular, automated tasks such as plant species identification, health status assessment from digital images, and seed classification and sorting constitute a substantial portion of AI-driven agricultural research and practice.

In recent years, research aimed at enhancing traditional agricultural practices through perception, information, and communication technologies has increased substantially. The integration of digital monitoring and computerized control systems has become particularly essential in controlled and indoor production environments, where activities such as irrigation, fertilization, and pesticide application are increasingly managed through automated systems. Numerous studies have focused on improving existing automation infrastructures and expanding their functional capabilities, with

artificial intelligence (AI)-based image recognition and classification systems representing a prominent and rapidly developing subgroup. Computer vision applications in agriculture have been widely investigated in areas such as crop growth monitoring [2–5], disease and pest detection [6–10], and product quality assessment [11, 12]. Alongside these domains, yield improvement and seed selection constitute distinct yet closely related research areas. As the primary biological input in crop production, seed quality, together with environmental conditions, plays a decisive role in determining yield potential.

According to standards established by the International Seed Testing Association (ISTA), seed quality classification is based on parameters such as genetic and physical purity, germination capacity, seed health, and moisture content [13]. To assess viability and germination performance, a range of chemical and physical techniques are employed, including protein electrophoresis, gas chromatography, high-performance liquid chromatography, tetrazolium testing, accelerated aging, and electrical conductivity measurements [14]. Although these methods can provide accurate and reliable results, they often require specialized equipment, expert personnel, and extended processing times, resulting in considerable operational costs. Similarly, conventional seed sorting and germination tests are labor-intensive, time-consuming, and may involve subjective evaluation [15, 16]. Consequently, the development of automated, objective, and cost-effective approaches for seed quality assessment has long been a significant research focus in agricultural engineering and smart farming systems [15].

In recent years, several studies have explored the use of image processing and machine learning techniques to improve the efficiency and objectivity of seed germination assessment. For example, Nehostan et al., developed a convolutional neural network (CNN)-based deep learning model, referred to as SeedNet, to predict germination from RGB images of seeds belonging to seven vegetable species. The proposed model achieved seed usability test accuracies ranging between 92% and 97% across different seed groups [17]. de Medeiros et al., investigated two fundamentally different methodological approaches to the seed germination problem. In their first study, soybean seeds were categorized into seven groups based on health and damage conditions. Linear Discriminant Analysis (LDA), Random Forest (RF), and Support Vector Machine (SVM) models were employed to classify root development status (germinated, weakly germinated, and not germinated) using images from 700 seeds. When evaluating differentiated seed groups such as healthy, damaged, and green seeds, the RF model trained with 10% of the dataset achieved an average classification accuracy of 95% across groups [18]. In their second study, the same research group utilized X-ray imaging and

Near-Infrared (NIR) spectroscopy data instead of RGB images. Images of *urochloa brizantha* seeds were classified as germinated or not germinated using LDA, Naïve Bayes (NB), RF, and SVM algorithms. Additionally, based on observations collected after twice the optimal germination period, a three-class classification (germinated, late germinated, and not germinated) was performed. The primary objective was to determine whether X-ray or NIR spectroscopy provided superior performance in germination prediction [19]. Genze et al., applied the Faster R-CNN algorithm to RGB images of corn, millet, and rye seeds germinated in petri dishes, classifying them into germinated and non-germinated categories [20].

The present study differs from the aforementioned works in several important aspects. First, it focuses on eggplant and okra seeds, which, to the best of our knowledge, have not previously been examined in image-based germination prediction studies. Furthermore, tomato seeds, one of the most widely produced vegetable crops worldwide, were included to provide a comparative and practically relevant reference. By incorporating both commonly cultivated and less-studied species, the study broadens the scope of seed germination modeling in literature. Second, unlike many previous studies that rely solely on RGB image data obtained under controlled laboratory conditions, this study integrates seed morphological characteristics with image-based features within a deep learning framework. Moreover, germination performance is evaluated based on seedling emergence from soil rather than laboratory-based germination tests conducted in petri dishes. This approach enhances the ecological validity of the findings, as soil emergence reflects not only physiological germination capacity but also the combined effects of biological vigor and environmental resistance under more realistic growing conditions. Finally, the proposed model's predictive performance is systematically compared with widely used state-of-the-art deep learning architectures, including ResNet50, VGG19, and EfficientNetB5, thereby providing comprehensive benchmark analysis.

The main contributions of this study can be summarized as follows:

1. Germination capacity prediction was performed for okra and eggplant seeds, species not previously addressed in image-based germination prediction studies, along with tomato seeds, one of the most extensively produced vegetable crops globally.
2. A deep learning-based model was developed that predicts germination with high accuracy by jointly utilizing morphological seed traits and RGB image data.
3. The predictive performance of the proposed model was comprehensively evaluated and compared

against established state-of-the-art deep learning architectures.

Related work

Computer-assisted studies aimed at increasing agricultural productivity primarily focus on minimizing the negative effects originating from seeds. These studies include classification of seeds, elimination of unhealthy seeds or disease-causing substances, determination of seed viability, and monitoring growth stages. Computer vision applications based on digital images provide an efficient and non-invasive alternative to manual observations. Such techniques have been widely used in food quality analysis and grain quality measurement [15]. For example, Kilic et al., used Artificial Neural Networks (ANN) trained with RGB images to predict damaged bean seeds, achieving an average classification accuracy of 90.6% on 371 samples [21]. To determine the quality of legumes, Mahajan et al., integrated image acquisition techniques with computer vision methods [22].

Computer vision has also been applied to monitoring crop growth stages. Sadeghi-Tehran et al. developed a system for automatic tracking of wheat heading and flowering stages using Support Vector Machines (SVM). Three stages of wheat heading (early, middle, and late) were evaluated, and prediction accuracies of 95.24%, 97.79%, and 99.59% were achieved, respectively [4]. Similarly, Iraj proposed a deep stacked sparse auto-encoders (DSSAEs) model for determining tomato seed quality directly from images, achieving a sensitivity of 83.2%, specificity of 96.50%, g-mean of 89.40%, and an overall accuracy of 95.5% [23]. Using two datasets comprising a total of 212 images, Bhargava and Bansal used SVM to automatically classify apples according to external quality, achieving accuracy scores of 96.81% and 93% [24]. de Medeiros et al., classified soybean seeds and identified their seedling potential using traditional machine learning methods, including SVM, Random Forest, and LDA, across three classes: strong seedlings, weak seedlings, and non-germinating seeds. Their approach achieved an average accuracy of 94% [18]. In a related study, de Medeiros et al., used X-ray images and machine learning methods (LDA, SVM, Random Forest, and Naive Bayes) to predict the germination and viability of forage grass seeds, achieving 85% accuracy for germination prediction and 62% for viability detection [19]. Li et al., used a Convolutional Neural Network (CNN) model for apple classification and quality grading from images, achieving an average accuracy of 95.33% on a dataset of 300 apples [12].

Beyond seed quality assessment, computer vision techniques have also been widely used for plant health monitoring and pest detection due to their potential to increase agricultural productivity. Naik et al., developed

an automatic system for evaluating iron deficiency chlorosis (IDC) in soybean plants using 5916 RGB images to detect plant stress levels, achieving an accuracy of 96% and implementing the system as a mobile application [25]. Maharlooeei et al., developed a method for the automatic detection and counting of aphids (*Aphis glycines*) on soybean leaves using MATLAB libraries, achieving an R^2 value of 0.96 for counting accuracy [8]. Liu and Chahl proposed a multispectral machine vision system based on UV imaging to detect invertebrate pests on green-leaved plants, introducing two models named Hypercube and Hyper Hue capable of detecting 12 common pest species [6]. Zhong et al., developed a Raspberry Pi-based system for counting and classifying flying insects using SVM and YOLO methods, achieving an average counting accuracy of 92.5% and a classification accuracy of 90.18% across six insect species [10]. Toseef and Khan proposed a mobile application that enables farmers to detect cotton and wheat diseases, achieving 99% accuracy in tests involving 100 disease types [26]. Zhao et al., proposed a deep learning model for detecting defects on the surface of soybean seeds, achieving an accuracy of 97.7% [27].

Seed germination is another important indicator of seed quality because it directly affects crop yield and productivity. Genze et al., investigated germination prediction for maize, rye, and pearl millet seeds using a dataset of 23,000 images and proposed an R-CNN-based model. The model was compared with ResNet50, ResNet101, Inception v2, and Inception-ResNet v2 architectures, achieving germination prediction accuracies of 97.9%, 94.2%, and 94.3%, respectively [20]. Colmer et al., developed a system called SeedGerm, consisting of both hardware and software components, which predicts seed

germination rates from PNG-format seed images using a U-Net CNN architecture [28]. More recently, Chen et al., evaluated the germination rate of corn seeds using RGB images collected from six different environments. A total of 181,250 images were analyzed, and nine CNN-based deep learning models were compared, with RTMDet achieving the highest performance [29].

Method

The study consists of three main phases: constructing the dataset, developing the model, and model evaluation. The schematic representation of all steps of the study is given in Fig. 1.

Construction of dataset

In the study, two separate datasets were created by capturing RGB images of seeds and extracting morphological traits of seeds. Okra (*Abelmoschus esculentus*), eggplant (*Solanum melongena*), and tomato (*Solanum lycopersicum*) seeds were obtained from the Ministry of Agriculture and Forestry for the 2023 harvest season. The seeds used in the study were not hybrid or subjected to chemical seed coating processes. RGB images of 405 seeds from each species were obtained from three different sources. The imaging process utilized a Celestron 2MP Digital Microscope, a Canon Mark II Digital Camera equipped with an EF-S 18–135 mm f/3.5–5.6 IS USM lens mounted on a tripod at a distance of 17 cm, and a Samsung scanner with a resolution of 600×600dpi. To mitigate the effects of shadows and lighting variation, a 6500 K color temperature ring light was employed for digital camera images. Preprocessing steps included cropping to seed bounding boxes and resizing to 460×460 pixels. A total

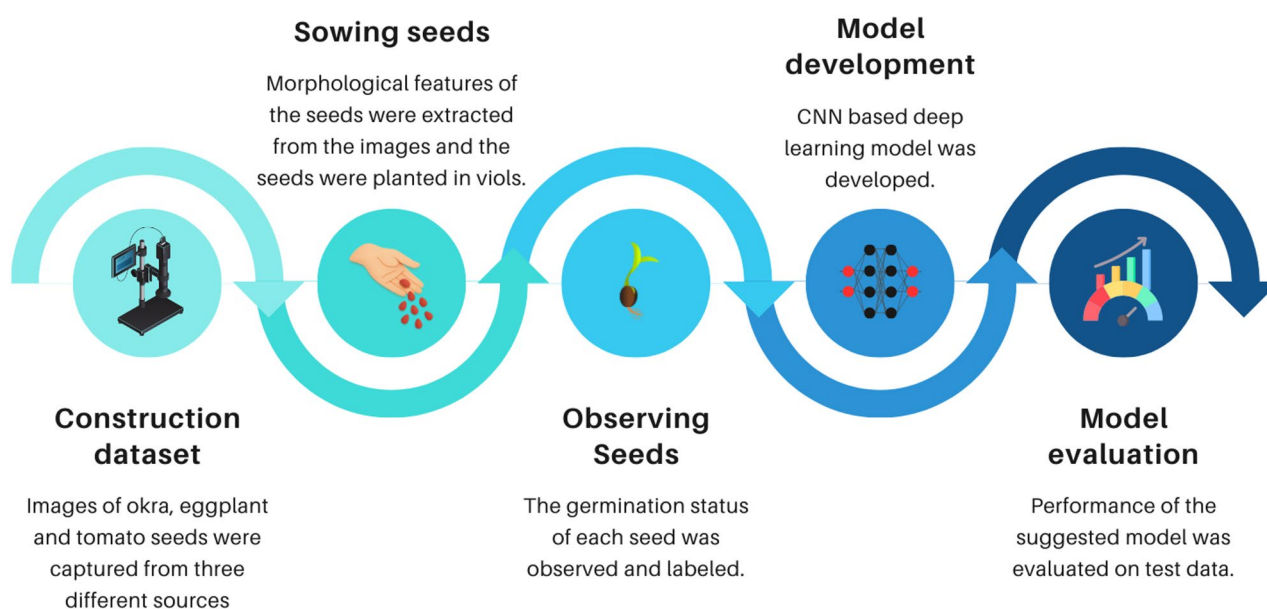
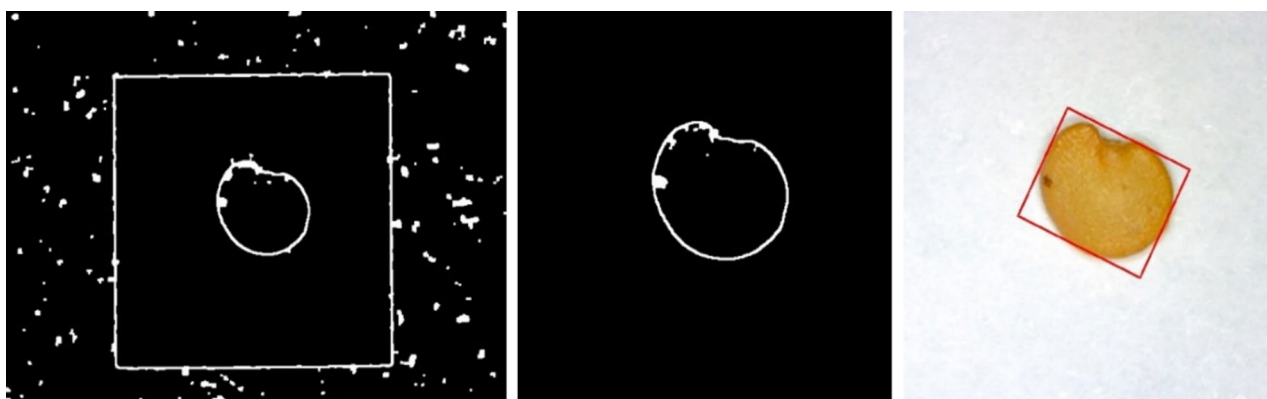


Fig. 1 Flow chart of the study

Table 1 Summary of seed image sources, quantities, average germination time, and performance allocation

Species	No. of Seeds	Digital Microscope	Digital Camera	Scanner	Total Images	Avg. Germination Time (day)	Train	Val	Test
Okra	405	405	405	405	1215	13	778	194	243
Eggplant	405	405	405	405	1215	14	778	194	243
Tomato	405	405	405	405	1215	12	778	194	243

**Fig. 2** Digital microscope images with 1 × 1 cm template (okra, eggplant, tomato)**Fig. 3** Extraction of morphological traits (eggplant seed)**Table 2** Seed morphological traits table (Cir. = circumference, CHA = Convex Hull Area)

	Area (mm ²)	Cir.(mm)	Width(mm)	Length (mm)	CHA	Solidity	W/H
Mean	22.82	17	4.71	5.49	23.63	0.96	0.86
Std.	12.33	5.94	1.27	1.36	12.39	0.04	0.08
Min.	0.792	8.63	2.48	3.24	7.87	0.02	0.54
Max.	56.28	89.94	7.94	9.43	58.03	0.99	0.99

of 3645 seed images were captured. Detailed information regarding the acquisition sources of the seed images, the associated seed characteristics, and the number of samples used in the training, validation, and testing phases are presented in Table 1.

Each seed was first placed on a created 1 cm × 1 cm template, and RGB images were captured at 640 × 480 pixel resolution using a digital microscope. The template and seed images are shown in Fig. 2.

The captured RGB images have been converted to black and white. Erosion and dilation operations were performed to minimize noise in the images. Subsequently, the object contours in the images were detected. Using the 1 cm × 1 cm square in the template as a reference, the

pixel-to-mm conversion was calculated, and the images were cropped to 460 × 460 pixels centered on the square. This allowed for seven different morphological traits of the seeds (width, length, area, perimeter, convex, solidity, width/length ratio) to be automatically extracted from the images. The sequence of operations performed is shown in Fig. 3.

The data on morphological traits of seeds are given in Table 2.

Later, images of each seed were taken at a resolution of 5472 × 3648 pixels using a digital camera from a distance of 17.5 cm, at an angle of 80° for okra and 90° for other seeds. Similarly, images of each seed were captured using a scanner. All images were resized to 460 × 460 pixels

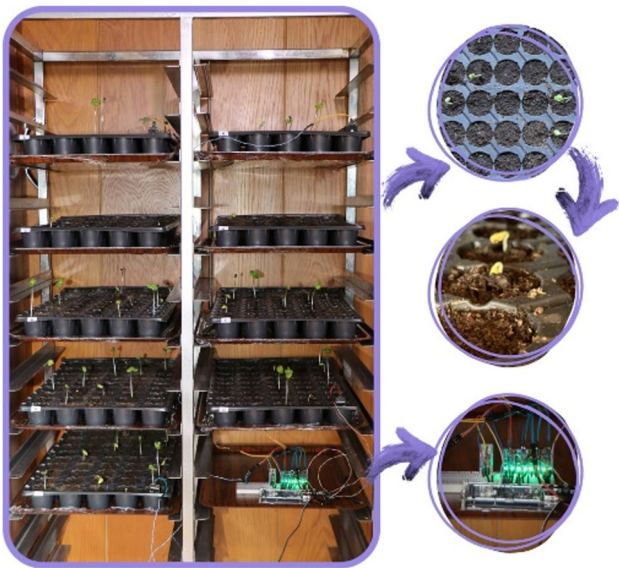


Fig. 4 Germination ambient

Table 3 Soil physicochemical properties

Chemical Data	Values	Nutrients	Values
pH value (H ₂ O, v/v1:2.5)	5,5	Nitrogen (mg N/l)	42
Salt level (g/l, v/v 1:3.6)	0,3	Phosphorus (mg P ₂ O ₅ /l)	30
		Potassium (mg K ₂ O/l)	54
		Magnesium (mg Mg/l)	80

with the seeds positioned at the center, ensuring that all images had the same resolution. Each image of seed was labeled. Not every seed that germinates in a laboratory environment can be expected to emerge above the soil surface in a real-life scenario. Following seed germination, the emergence of the seedling above the soil surface represents a critical developmental stage, as it enables the initiation of photosynthesis, the establishment of water-air balance, the overcoming of mechanical barriers, and the perception of environmental signals [30]. This stage

is also vital because it marks the transition from reliance on stored energy reserves to the onset of autonomous energy production through photosynthesis [31]. For this reason, the present study considered not only seed germination but also seedling emergence above the soil surface as a primary criterion. Each image captured and labeled seed was sowed in 45-cell tray to create real soil conditions. The soil moisture in the trays and the ambient temperature was monitored using a microcontroller (Arduino). The germination ambient is shown in Fig. 4.

To provide the most suitable soil structure for seed germination, a mixture of white sod peat (1–7 mm), white peat (0–5 mm) and 15% fine perlite was used in the seedling trays. Chemical and nutrient data related to the soil are given in Table 3.

Each seed seedling that emerged on the soil surface was classified as germinated, while the rest were classified as non-germinated. The germination rates were as follows: okra seeds, 54.32%; eggplant seeds, 80%; and tomato seeds, 63.95%. Thus, a dataset containing 3645 seed images and morphological traits from three different sources belonging to three species was created. Figure 5 shows the relationships between the morphological traits of the seeds.

To ensure data integrity and accurate labeling, a master’s-level agronomy specialist, independent of the researcher, conducted a cross-validation of the germination data. All procedures were carried out under the supervision of the researcher’s academic advisor, a professor.

Model development

The use of multi-layered structures has enabled deep learning models to be used in all areas of life. In applications where image-based deep learning methods are used, CNN, a special type of artificial neural network structure, is frequently utilized [32]. Thanks to the

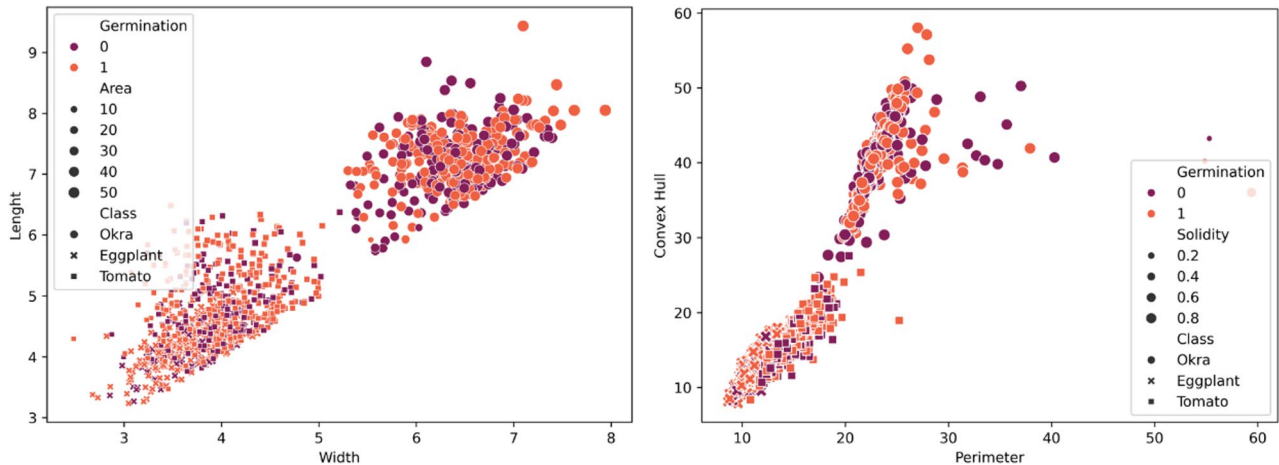


Fig. 5 Seed morphological traits

ability of CNN structures to extract important and automatic features from images, preprocessing on images is not necessary. What differentiates CNN from other ANN structures is that it has a convolution layer where features are extracted, as well as pooling layers where the extracted features are aggregated and the network dimension is reduced. The most important feature of CNN structures is the convolution process, which is the process of extracting features from images. The convolution operation is the process of obtaining new values by moving a specific kernel or filter over image channels (Eq-1).

$$F(i, j) = (I * K)(i, j) = \sum_m \sum_n I(m, n) K(i - m, j - n) \quad (1)$$

F , I and K represent the feature matrix, the inputs and the kernel/filter, respectively. The $*$ symbol represents the convolution operation, m, n represent the filter elements, and i, j represent the output pixels.

In the convolution process, the stride parameter determines the step size and direction in which the filter moves across the input matrix (whether the input image or feature map). The stride significantly influences the dimensions of the resulting output feature map. Additionally, padding is employed alongside stride to control the size of the output feature maps and to allow the convolutional filters to access the edge pixels of the input. Stride and padding operations are given Eq-2.

$$H_o = \left(\frac{F_h - f + 2p}{s} + 1 \right), W_o = \left(\frac{F_w - f + 2p}{s} + 1 \right) \quad (2)$$

F_h , F_w and represent the height, width of the feature matrix, respectively, f represents the size of the filter, s represents the stride, and p represents the padding. W_o , H_o are the width and height of the output feature map, respectively.

After the convolution operation, a nonlinear activation function such as ReLU (Rectified Linear Unit) is typically applied. ReLU is the most commonly used activation function in NN structures. This enables the model to learn nonlinear complex patterns. The ReLU function is given Eq-3.

$$ReLU(x) = \max(0, x) \quad (3)$$

Using numerous filters for feature extraction brings along the problem of increasing the data to be calculated. To reduce computational and resource costs and make the model more feasible, it is necessary to summarize the extracted features or take the most important ones. Thus, both the most important features are obtained, and the data dimension is reduced. Pooling layer is used to perform this operation. The pooling layer preserves the most

important features while reducing the CNN dimension. The operation of the max pooling layer is given in Eq-4.

$$y_{i,j} = \max(x_{i,j} | i \in \{i, i+1\}, j \in \{j, j+1\}) \quad (4)$$

$y_{i,j}$ is the value at position in the output matrix, $x_{i,j}$ represents the pixel values in the input matrix within the pooling window, and $\max$ function selects the maximum value from the window.

In the model proposed in this study, ReLU was used between layers, while the sigmoid function was utilized in the final layer. The proposed model is given in Fig. 6.

The proposed model is composed of three main sub-blocks. The first sub-block is a convolutional block (CB), which is responsible for hierarchical feature learning from RGB images using convolutional layers. The increasing depth and complexity of CNN architectures do not necessarily yield proportional improvements in performance. The growth in the number of parameters, particularly when training data are limited, may weaken the model's generalization capability [33]. The highly parameterized nature of deep neural networks, while contributing to improved model performance, often leads to diminished generalization capacity [34]. Furthermore, an excessive number of parameters have been shown to increase computational costs and adversely affect inference time [35]. To preserve generalization ability and reduce computational overhead, a two-layer CNN architecture was selected as the optimal experimental configuration. The second sub-block is a feature extraction module (FE) that incorporates morphological traits to enhance the representational capacity of the model. The third sub-block uses a deep neural network (DNN) to perform the final classification based on the combined feature representations, along with seven morphological traits of the seeds and the seed class information, obtained from the previous stages. Thus, a lightweight model was developed for germination capacity prediction. The detailed parameters of the proposed model are provided in Table 4. In model training and test phases, the Adam optimizer was employed during training. The initial learning rate for all models was set to 1×10^{-3} , and after a patience of five epochs, the learning rate was reduced by a factor of 0.2. Minimum learning rate was set 10^{-5} .

The computational characteristics of the proposed model, including the number of trainable parameters, total training time, and average inference time per sample, are compared with those of other well-known models and summarized in Table 5.

Since the created dataset is not a balanced dataset, ROC AUC score and weighted F1 metrics were used instead of accuracy in measuring the model's performance. To calculate the F1 score, precision and recall

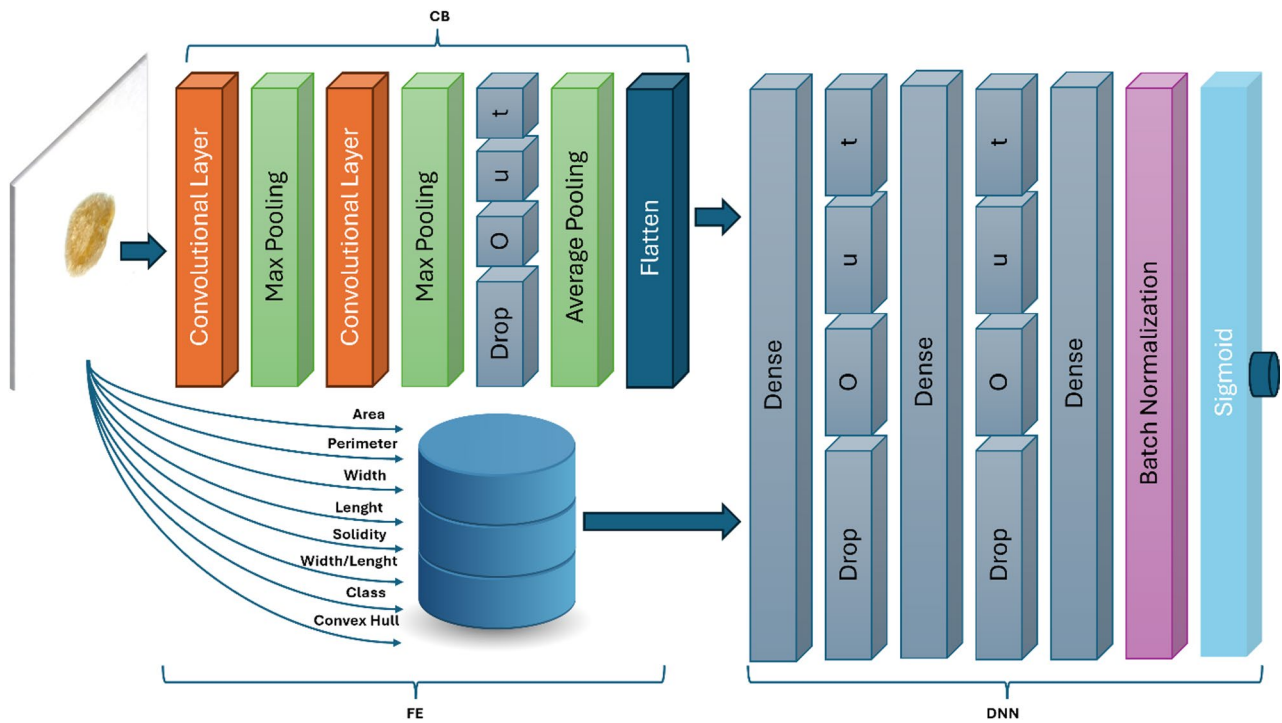


Fig. 6 Proposed model with its sub-blocks

Table 4 Proposed model hyperparameters

Layer Operation	Input Size	Output Size	Activation Functions	Filter Size	Filters/Rate
Inputs RGB	460 × 460 × 3	460 × 460 × 3	-	-	-
Convolution 2D	460 × 460 × 3	458 × 458 × 32	ReLU	3 × 3	32
Maximum Pooling	458 × 458 × 32	114 × 114 × 32	-	4 × 4	-
Convolution 2D	114 × 114 × 32	112 × 112 × 16	ReLU	3 × 3	16
Maximum Pooling	112 × 112 × 16	37 × 37 × 16	-	3 × 3	-
Dropout	37 × 37 × 16	37 × 37 × 16	-	-	0,5
Average Pooling	37 × 37 × 16	18 × 18 × 16	-	2 × 2	-
Flatten	18 × 18 × 16	5184	-	-	-
Concatenated	5184	5192	-	-	-
Dense	5192	180	ReLU	-	-
Dropout	180	180	-	-	0,2
Dense	180	120	ReLU	-	-
Dropout	120	120	-	-	0,2
Dense	120	60	ReLU	-	-
Batch Normalization	60	60	-	-	0,2
Dense	60	1	Sigmoid	-	-

need to be calculated. The calculations of precision, recall, F1, weighted F1 and ROC AUC metrics are given in Eqs. 5–10, respectively.

$$Precision = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (5)$$

$$Recall = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (6)$$

$$\text{False Positive Rate (FPR)} = \frac{\text{False Positives (FP)}}{\text{False Positives (FP)} + \text{True Negatives (TN)}} \quad (7)$$

$$F1 = 2x \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

$$weightedF1 = \frac{\sum_{i=1}^N w_i x F1_i}{\sum_{i=1}^N w_i} \quad (9)$$

Table 5 Comparison of complexity of the proposed model with well-known models

Models	Trainable Parameters	Total Parameters	Avg. Training Time per Epoch (s)	Avg. Inference Time per Sample (ms)
VGG19	20,025,921	20,026,945	370.17	327
ResNET50	23,540,737	23,597,953	176.11	315
EfficientNETB5	28,460,142	28,523,768	392.45	422
Proposed Model	969,421	969,541	35.22	104

Table 6 Performance evaluation of sub-blocks of the proposed model

Models	Avg. Accuracy	Avg. weighted F1-score	Avg. ROC AUC	Standard Deviation*
CB + DNN	0.85	0.85	0.83	0.021
FE + DNN	0.74	0.75	0.74	0.039
Proposed Model	0.95	0.95	0.94	0.020

*The weighted F1-score standard deviation was reported

$$ROC\ AUC = \int_0^1 Recall(FPR) d(FPR) \quad (10)$$

In Eq-9, w_i is the ratio of the number of examples in class i in the data set to the total number of examples.

Model training and testing were conducted on a computer with an Intel Xeon E-2246G processor and 32GB of memory.

Results and discussion

To determine whether seed morphological characteristics affect germination capacity prediction, the three sub-modules of the proposed model were tested separately. The first sub-module (CB + DNN) was trained using solely the RGB images of all seeds, and its germination prediction performance was measured. In a similar manner, the other sub-module (FE + DNN) was trained using the morphological traits of all seeds, and its prediction performance was measured. Subsequently, the proposed model was trained using both seed images and seed morphological traits, and its prediction performance was measured. Performance measurements were conducted using 5-fold stratified cross-validation for each sub-block independently. The prediction performance of the sub-blocks and the model is presented in Table 6.

Pre-trained models were used to compare the germination prediction performance of state-of-the-art models frequently used in image-based studies, such as ResNET50, EfficientNetB5, and VGG19, with the proposed model. The pre-trained weights of the models on the ImageNet database were used, and the last two layers were trained. All models were tested over 100 epochs with different batch sizes. Most studies on image-based deep learning models have indicated that using small batch sizes and slow learning rates positively affects the success of deep learning models [36, 37]. Similarly, in the current study, the highest accuracy rate was achieved

when the batch size was 32. Training and testing data were implemented based on these values. In order to meet the high data demands of deep learning models, during the training phase, each seed image was augmented fivefold through the application of random rotation, flipping, and contrast adjustment techniques as part of the preprocessing pipeline. Since a data set created with images of each seed from 3 different sources was used, the seeds were separated into training and testing by separating them on a seed basis to ensure that the images of the seed in the training dataset not in the test dataset or validation dataset. The dataset was randomly divided into 80% training and 20% testing, and training was performed under 5-fold stratified cross validation. In each fold, the model was trained with 80% of the training data and the model performance was validated with 20% of the training data. The scores after each fold were recorded and the average performance score of the model after each epoch were calculated. Average training and loss scores of the models are given in Fig. 7.

When examining the training graphs of the models, it can be observed that while other models show almost stable accuracy after 50 epochs, the proposed model reaches its highest accuracy after 90 epochs. Similarly, when loss is examined, well-known models complete their learning phase after 50 epochs, whereas the proposed model achieves its highest performance success after 90 epochs. The high number of parameters in well-known models increases the accuracy in each epoch. While this allows for higher predictions with fewer epochs, it also limits the generalization capability of the models and causes the emergence of overfitting problems [38]. The average validation performance results of the models after 5-fold stratified cross validation are provided in Fig. 8.

The validation accuracy of the models shows that while the prediction success of well-known models stabilizes after 50 epochs in parallel with the training, the performance of the proposed model continues to increase up to 90 epochs. As shown in the validation loss graph, the loss values of well-known models remain constant between 20 and 30 epochs. However, the loss value of the proposed model decreases during this time. It is understood that seed images alone are not sufficient for germination prediction. In the proposed model, in addition to RGB images of seeds, morphological features of seeds were also used, which increased the model's success in germination prediction. Datasets created in germination

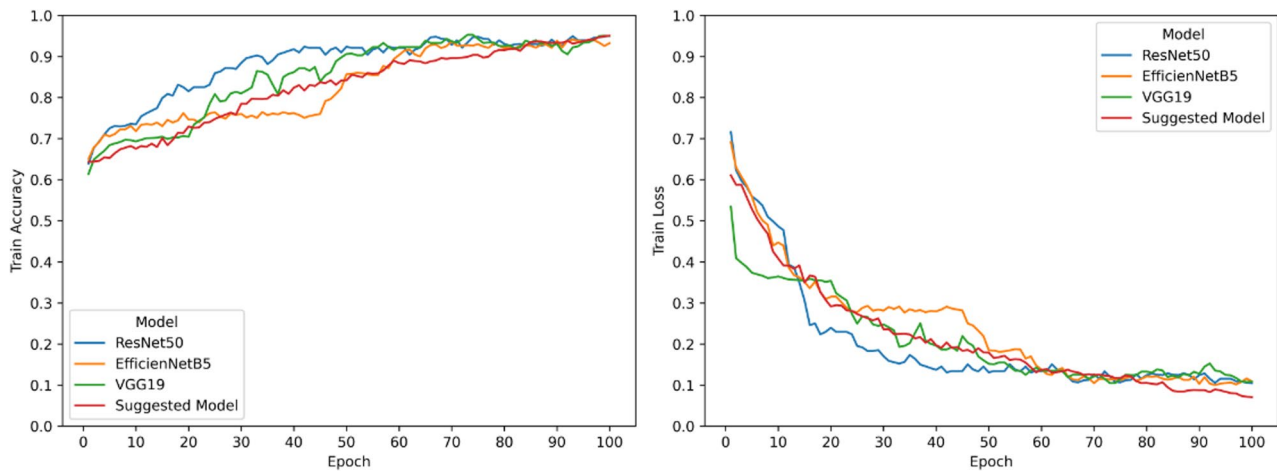


Fig. 7 Models' average training accuracy and loss

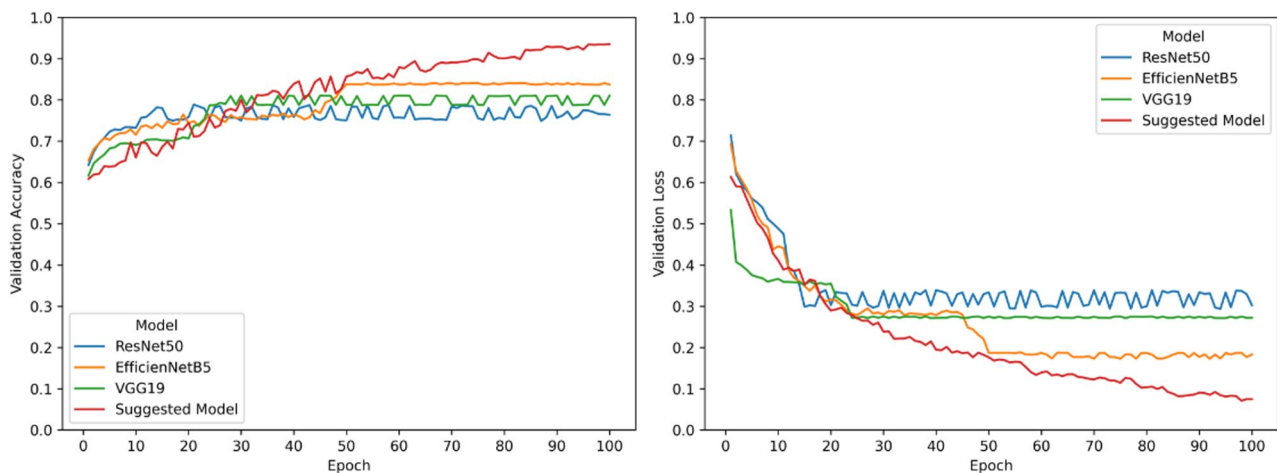


Fig. 8 Models' average validation accuracy and loss

studies on seed images are imbalanced datasets. Examining the performance of models on imbalanced datasets through correct metrics helps to evaluate the true performance of the models [39]. Since the dataset created in the study is not a balanced dataset, weighted F1 score and ROC AUC score were used as the basis for model performance evaluation. Both scores were chosen for model performance evaluation because they support classes that are less represented in the dataset. The averages of the models' 5-fold stratified cross-validation training performances, along with their test performances, are presented in Table 7.

As shown in Table 7, although all models achieved relatively strong performance during the training phase, VGG19 and the proposed model demonstrated the highest training success. However, in the testing phase, only the proposed model maintained a comparable level of performance, whereas other established deep architectures exhibited a substantial decrease in F1 scores.

The observed decline in testing performance for high-capacity CNN models can be attributed to their structural complexity and data requirements. Deep architectures such as ResNet50 and EfficientNetB5 contain a large number of parameters and were originally designed to be trained on very large-scale datasets. While their high representational capacity enables them to model highly complex patterns, this same capacity may hinder generalization when the available training data are limited or domain-specific. Under "small sample size" conditions, excessively complex CNN structures may align too closely with training data patterns, leading to overfitting and reduced ability to capture broader data distributions. This phenomenon is consistent with findings in the literature demonstrating that structural complexity can negatively affect generalization when dataset size is insufficient, and that architectures better matched to the scale of the data tend to achieve improved test performance [40].

Table 7 5-fold stratified cross-validation and test performance of the models

Phase	Seed	Metric	Proposed Model	VGG19	ResNet50	EfficientNetB5
TRAIN	OKRA	0	precision	0.86	0.83	0.84
			recall	0.89	0.89	0.89
			f1	0.87	0.86	0.86
		1	precision	0.93	0.93	0.93
			recall	0.91	0.89	0.90
			f1	0.92	0.91	0.91
		weighted f1		0.90	0.89	0.91
		ROC AUC Score		0.90	0.88	0.90
		Standard Deviation		0.031	0.019	0.024
	EGGPLANT	0	Precision	0.96	0.96	0.96
			Recall	0.94	0.94	0.94
			f1	0.95	0.96	0.95
		1	Precision	0.96	0.97	0.96
			Recall	0.97	0.97	0.97
			f1	0.97	0.97	0.96
		weighted f1		0.96	0.97	0.95
		ROC AUC Score		0.95	0.97	0.95
		Standard Deviation		0.018	0.022	0.016
	TOMATO	0	precision	0.94	0.92	0.93
			recall	0.91	0.92	0.90
			f1	0.93	0.92	0.91
		1	precision	0.95	0.95	0.94
			recall	0.97	0.95	0.96
			f1	0.96	0.95	0.95
		weighted f1		0.95	0.94	0.94
		ROC AUC Score		0.94	0.94	0.94
		Standard Deviation		0.024	0.017	0.021
	OKRA+EGGPLANT+TOMATO	0	precision	0.92	0.93	0.90
			recall	0.95	0.95	0.88
			f1	0.94	0.94	0.89
		1	precision	0.96	0.95	0.95
			recall	0.95	0.95	0.97
			f1	0.95	0.95	0.95
		weighted f1		0.95	0.95	0.94
		ROC AUC Score		0.94	0.95	0.91
		Standard Deviation		0.020	0.018	0.022

Table 7 (continued)

Phase	Seed	Metric	Proposed Model	VGG19	ResNet50	EfficientNetB5
TEST	OKRA	0	precision	0.84	0.62	0.57
			recall	0.82	0.67	0.63
			f1	0.83	0.64	0.60
		1	precision	0.85	0.78	0.75
			recall	0.87	0.74	0.70
			f1	0.86	0.76	0.72
		weighted f1		0.85	0.72	0.68
		ROC AUC Score		0.84	0.70	0.66
	EGGPLANT	0	precision	0.85	0.70	0.66
			recall	0.88	0.80	0.73
			f1	0.87	0.75	0.69
		1	precision	0.96	0.86	0.82
			recall	0.95	0.79	0.77
			f1	0.96	0.83	0.79
		weighted f1		0.93	0.80	0.75
		ROC AUC Score		0.91	0.79	0.75
	TOMATO	0	precision	0.86	0.64	0.61
			recall	0.87	0.77	0.71
			f1	0.86	0.70	0.65
		1	precision	0.92	0.84	0.80
			recall	0.92	0.73	0.71
			f1	0.92	0.78	0.75
		weighted f1		0.90	0.75	0.71
		ROC AUC Score		0.89	0.75	0.71
	OKRA+EGGPLANT+TOMATO	0	precision	0.89	0.73	0.65
			recall	0.92	0.81	0.71
			f1	0.90	0.76	0.68
		1	precision	0.95	0.80	0.76
			recall	0.94	0.87	0.79
			f1	0.95	0.83	0.77
		weighted f1		0.93	0.81	0.74
		ROC AUC Score		0.92	0.81	0.76

Values given in bold represent the highest performance values.

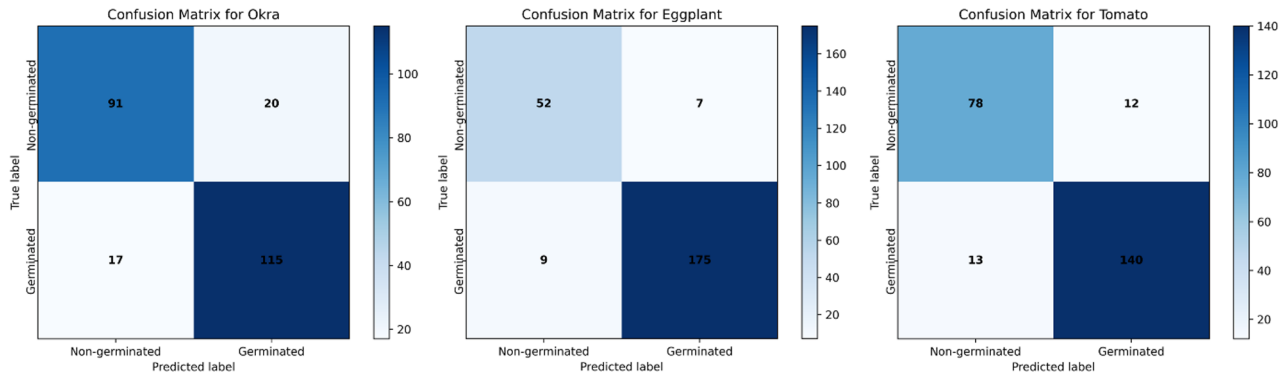


Fig. 9 Class-wise prediction performance of the proposed model

In contrast, the proposed model appears to be more appropriately aligned with the size and characteristics of the germination dataset. By providing a more suitable balance between architectural complexity and dataset scale, it is able to learn discriminative features

without overfitting. This architectural compatibility likely explains its superior generalization performance in germination prediction.

The test confusion matrices are presented in Fig. 9. These matrices further illustrate the class-wise prediction

performance of the proposed model and confirm its consistent and balanced behavior across different classes during the testing phase.

To assess the differences in performance among the models are statistically significant, a non-parametric Friedman test was applied to the results of 5-fold cross-validation and test evaluations. The analysis revealed that the differences between the models were found to be statistically significant (training scores; $\chi^2(3) = 10.67$, $p = 0.013$ and test scores; $\chi^2(3) = 11.09$, $p = 0.011$; $p < 0.05$).

One of the factors influencing the predictive performance of the models is the relatively low germination rate observed in the okra seeds used in this study. This outcome is not attributed to inadequate experimental conditions; rather, it may be associated with batch-specific seed viability characteristics and natural biological variability. In addition, the absence of chemical coating treatments, commonly applied to enhance moisture retention and protect seeds from external stress factors, may have contributed to increased variability in germination outcomes. Such inherent variability in the dataset increases the complexity of the classification task, making robust prediction more challenging for all models.

Nevertheless, the low germination rate further underscores the practical significance of this study. In many similar investigations on seed germination, seeds are subjected to chemical coating treatments or experiments are conducted under controlled, soilless laboratory conditions to maximize germination success [19, 28, 29]. In contrast, the present study evaluates germination under more natural and untreated conditions, without artificial enhancement of seed performance. From this perspective, the study differs methodologically from prior research and offers a framework that is more representative of real-world agricultural scenarios, thereby increasing its practical applicability.

Conclusions

Increasing agricultural productivity depends fundamentally on the selection of high-quality seeds. In this study, a lightweight deep learning model was developed within the framework of smart agriculture to automate the seed quality assessment process, which is traditionally labor-intensive and costly. By integrating morphological seed traits with RGB image data, the proposed model achieved a weighted F1-score of 0.93 in predicting germination capacity across three different seed types. These findings demonstrate that combining structural seed characteristics with image-based features can provide a robust and efficient approach for germination prediction.

The results indicate that the proposed framework offers a practical and scalable solution for early-stage seed quality evaluation, with potential applications in automated seed sorting, quality control, and decision-support

systems in agricultural production chains. The lightweight architecture further supports its potential integration into real-time or resource-constrained environments, contributing to the broader implementation of intelligent and data-driven agricultural practices.

Beyond its immediate predictive performance, this study demonstrates the feasibility of integrating multi-source feature representations in seed analysis tasks and provides a methodological foundation for future research in image-based agricultural modeling. Notably, to the best of our knowledge, image-based germination prediction for okra and eggplant seeds has not been previously addressed in the literature. By extending deep learning-based germination analysis to these relatively unexplored seed types, the study contributes a novel application domain and enhances the originality of the proposed framework. Expanding the approach to larger and more diverse datasets, incorporating additional imaging modalities, and adapting the framework to different crop species may further increase its applicability and impact in sustainable and precision agriculture systems.

Limitations

The fact that the model was developed and tested using only three different seed types necessitates cautious interpretation regarding the generalizability of the proposed approach to other plant species and seed groups with distinct morphological traits. Since seed traits such as size, shape, surface texture, and color distribution may vary substantially across species, the performance of the model on different seed groups should be further validated.

In addition, although image data were obtained from three different devices (microscope, camera, and scanner), a detailed and comparative device-specific performance analysis was not comprehensively presented in this study. These devices differ fundamentally in their imaging principles, spectral sensitivity ranges, and spatial resolutions. Such differences may influence image properties including contrast, color fidelity, and level of detail, thereby potentially affecting the predictive accuracy of the model. Therefore, the impact of different imaging devices on model performance should be statistically evaluated through device-based analyses, and cross-device generalizability should be systematically examined in future studies. Apart from these considerations, the potential impact of differences in image acquisition angles on prediction performance should be further investigated. Additionally, beyond RGB-based germination prediction, future studies may also investigate alternative color spaces (e.g., HSV, CMY, HSI) and advanced imaging techniques such as X-ray and infrared modalities to improve predictive robustness and cross-modality generalization.

Authors' contributions

This study was produced from the doctoral dissertation prepared under the supervision of the second author.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations**Ethical approval**

The study does not involve human/animal participants, ethical approval from an institution is not required. However, the researcher confirms compliance with the ethical principles of scientific publication.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

Consent to participate

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